# MISTAKING THE MAP FOR THE TERRITORY
## A Systems-Theoretical Response to the Platonic Representation Hypothesis

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**Abstract**

Huh, Cheung, Wang, and Isola (2024) argue that representations learned by AI models of different architectures, objectives, and modalities are converging on a "platonic representation": a shared statistical model of an underlying, mind-independent reality $Z$. We read this claim through Niklas Luhmann's operational constructivism and argue that it mistakes structural coupling for correspondence. No system—biological, social, or artificial—has operational access to a world-in-itself; it accesses only its own, already-selected environment. Applied to foundation models, this yields a claim stronger than the usual objection that training data carries sociological bias: convergence does not approach reality to a *degree* that more data could close, because the gap between model and $Z$ is constitutive, not contingent. We sharpen this argument by asking whether large models are autopoietic systems in Luhmann's sense at all, and argue that their very convergence is evidence that they are not: convergence is what a trivial, input-output machine does when its environment is stable, not what an autopoietic system does. We test the resulting account against the strongest piece of concrete evidence in Huh et al. (2024)—the convergence of color representations onto CIELAB space—and derive an empirical prediction that diverges from theirs: scaling should reduce disagreement with the statistical redundancy of the training corpus, not with reality, leaving rare-but-true, low-redundancy claims vulnerable to hallucination regardless of scale. We close by noting that the Platonic representation hypothesis is one exhibit among several of the same underlying move in and around machine learning research, from the Anna Karenina scenario to claims of universality in interpretability and brain-model alignment.

## 1 Introduction

The rapid evolution of artificial intelligence (AI) has recently yielded a phenomenon as striking as it is provocative: convergence. Across widely divergent architectures, training objectives, and data modalities, deep learning models are beginning to represent the world in increasingly isomorphic ways. Vision transformers, large language models, and diffusion models—once distinct in their internal topology—are now demonstrating a measurable alignment in their representation spaces. This observation has led Huh et al. (2024) to propose the *Platonic representation hypothesis (PRH)*. Their argument is bold, interesting, and intuitively appealing: if different systems, observing different data, converge upon a similar internal structure, they must be discovering a shared, underlying reality—the "shadows on the wall" of Plato's cave are finally revealing the objects casting them.

By aligning their hypothesis with the tradition of *convergent realism*, Huh et al. (2024) echo foundational defenses of the idea that inquiry steadily approaches an objective truth (Doppelt, 2007; Putnam, 1982). Central to this view is what Hilary Putnam famously termed the *no miracles argument* (NMA): the notion that the predictive success of our theories would be a miracle if they did not, at least approximately, map onto a mind-independent reality. The PRH conjectures that AI is gradually solving the inverse problem of perception to reveal an objective ontology driven by the statistical structure of the physical world itself.

This paper argues that the inference is unwarranted, but not for the reason usually given against it. The familiar objection—that training corpora carry sociological bias and are therefore an imperfect sample of reality—is one Huh et al. (2024) already anticipate and partly concede. Our claim is stronger and of a different kind: convergence is not evidence that models are approaching a mind-independent reality $Z$, nor, read the other way, is it evidence that models are autonomously constructing a world of their own. Read through Niklas Luhmann's operational constructivism, convergence is what happens when structurally closed systems become tightly coupled to the same environment—in this case, the sedimented redundancy of human communication, not the physical world that communication is about. This is a claim about the structure of observation as such, not about a correctable defect in a dataset: no amount of additional data closes the gap, because the gap was never a matter of degree. We push this further than a purely conceptual reading by asking whether large models are *systems* in Luhmann's technical sense at all, and we argue that the very fact of convergence is evidence that, in the strict sense, they are not—which sharpens rather than weakens the argument against the PRH.

While the empirical observation of convergence is rigorous, the metaphysical interpretation offered by the PRH serves as a welcomed provocation. By framing this convergence as an approximation of an independent physical reality, Huh et al. (2024) draw our attention to core questions regarding the interface between data-driven AI and social, living, and psychic systems. This paper suggests, however, that the PRH may represent the computational apotheosis of what Rorty (1979) critiqued as the "mirror of nature"—the ancient metaphor that the goal of inquiry is to reflect an external world as accurately as possible. The PRH invites us to examine whether the emerging consensus among artificial observers reflects the essence of the observed, or whether it indicates a *systemic resonance* emerging from the history of communicative selections.

We treat the PRH as the anchor case of this paper, not as an isolated target. It earns that role because it is, among its kind, unusually explicit: it gives the underlying metaphysical wager a name, a mathematical argument, and a mechanism. The same wager recurs, less explicitly, throughout adjacent research: in the "Anna Karenina scenario", the observation that well-performing networks represent the world in the same way while poorly-performing ones fail idiosyncratically (Bansal, Nakkiran, & Barak, 2021); in claims of *universality* in mechanistic interpretability, where independently trained vision models are found to learn "the same" curve detectors and high-low frequency detectors, treated as evidence that networks discover shared algorithmic structure in their input (Olah et al., 2020); and in the neuroAI literature on brain-model alignment, where the representations of artificial and biological visual systems are compared under the assumption that both are approximating the same visual world (Schrimpf et al., 2018; Yamins et al., 2014). Each of these recruits a version of the same inference—shared structure implies a shared, mind-independent object of representation. We develop our argument against the PRH in full because it states this inference most openly, and we return briefly, in the conclusion, to what the argument implies for its less explicit relatives; a systematic treatment of that broader literature is left to future work.

In this paper, we challenge the representationalist assumptions of the PRH through the lens of *operational constructivism*. We build upon the view that large foundational models are not passive sensors of physics, but rather *cultural and social technologies* (Bender, Gebru, McMillan-Major, & Shmitchell, 2021; Esposito, 2022; Farrell, Gopnik, Shalizi, & Evans, 2025; Klein et al., 2025; Sorensen et al., 2024; Zönnchen, Dzhimova, & Socher, 2025). We argue that the convergence observed in modern AI systems is not a convergence on "nature," but rather a convergence on a statistical model of sedimented communication seen as a contingent, socially patterned manifestation of meaning (*Sinn*) (Luhmann, 1997, p. 42).

Drawing on social systems theory (Luhmann, 1988) and second-order cybernetics (von Foerster, 1974), we posit that cognition presupposes a system-environment boundary, implying that meaning is strictly an internal construction. No cognitive system—whether biological, social, or artificial—possesses direct access to the "world-in-itself"; it accesses only its structurally coupled environment, which is always already a selection performed by the system—an order produced from noise. From this perspective, the "Platonic" manifold discovered by AI is not the ground truth of the universe, but the most viable statistical average of human linguistic and visual distinctions.

The remainder of the paper proceeds as follows. Section 2 reconstructs the PRH on its own terms, including the hedges its authors already build into it. Section 3 situates the PRH within, and against, the trajectory of twentieth-century realism debates. Section 4 introduces operational constructivism and, in particular, shows why Putnam's no-miracles argument dissolves rather than needing to be refuted once science is described as a truth-coded social system. Section 5 applies this apparatus to the PRH directly: it asks whether large models are systems in the relevant sense, sharpens the closure argument against the PRH's own concessions, rereads the PRH's color-cooccurrence experiment as a case study, and restates the difference between viability and validity. Section 6 derives a testable disagreement between the PRH's account of scaling and ours. Section 7 concludes.

## 2 A New Mirror of Nature

Before we deploy a constructivist critique, however, we must rigorously articulate the PRH on its own terms. The hypothesis is not merely a philosophical speculation; it is grounded in a series of robust empirical observations that demand explanation.

The core of the PRH rests on a phenomenon that is becoming increasingly difficult to ignore: convergent representational evolution. As deep learning models scale in parameters and data ingest, their internal geometries are becoming isomorphic, regardless of their architectural provenance.

Huh et al. (2024) quantify this using metrics such as *centered kernel alignment (CKA)* and Procrustes analysis. If we formalize the representation space of a model as a kernel function $K(x, x')$, the empirical data suggests that for any two sufficiently advanced models $M_1$ (e. g., a large language model) and $M_2$ (e. g., a vision transformer):

$$\text{Alignment}(K_{M_1}, K_{M_2}) \to 1 \quad \text{as} \quad \text{Scale} \to \infty \tag{1}$$

This convergence is non-trivial. $M_1$ and $M_2$ may be trained on disjoint datasets (text vs. pixels) and optimized for distinct objective functions (next-token prediction vs. contrastive loss). Yet, they arrive at a shared statistical topology. The PRH argues that this alignment is not accidental; it implies that the models are converging on a universal, optimal way of encoding information—a Platonic manifold.

To explain this mathematical convergence, PRH proponents invoke a causal realism rooted in physics. They posit the existence of a latent variable $Z$, representing the underlying physical state of the world. In this framework, the data modalities humans perceive and machines ingest—text, images, audio—are merely projection functions ($f$) of this reality. As the authors put it in their own statement of the hypothesis, explicitly invoking Plato's allegory of the cave:

> "Neural networks, trained with different objectives on different data and modalities, are converging to a shared statistical model of reality in their representation spaces." – (Huh et al., 2024, p. 1)

In this view, the world model is not constructed; it is recovered. When a vision model learns the vector for "cup" and a language model learns the embedding for "cup", they are solving the inverse problem of perception to triangulate the same physical object $Z_{cup}$. Convergence is thus interpreted as the gradual elimination of modal-specific noise, revealing the clean signal of physical law. The implicit claim is that as error rates approach zero, the representation approaches ontology.

Philosophically, the epistemological warrant for the PRH relies on a computational variation of Hilary Putnam's "no miracles argument" for *scientific realism* (Putnam, 1979). Putnam argued that the consistent success of scientific theories in predicting phenomena would be "miraculous" if those theories were not, at least approximately, true descriptions of an independent reality. His argument goes like this:

> "The positive argument for realism is that it is the only philosophy that doesn't make the success of science a miracle."

In other words: If our best scientific theories did not correctly describe the world, why should we expect them to be successful? On the other hand, since the truth of our best theories is an excellent explanation of their success, we should accept the realist hypothesis: our best scientific theories are true and constitute knowledge of the world (Sprenger, 2016).

Putnam argued that if science was just making things up or using "useful fictions", how could we explain that our planes fly, our GPS works, and our medicines cure diseases? It would be a statistical miracle if our theories were wrong about reality but still worked so perfectly. Therefore, scientific theories must be converging (getting closer and closer) on the actual truth of the universe. Therefore, we should accept that the terms in our theories (like "electrons" or "DNA") actually refer to real things, because that reference explains why the theories are successful.

This sounds totally reasonable even though it is an argument based on an empirical *observation* of science itself, i. e., if science operates under the true/false distinction it produces constantly true as well as false theories (which are often marked as true before). From a systems theoretical perspective, it seems trivial that science is successful in the production of truth. I come back to this point later.

The PRH extends this logic to neural networks: it would be arguably miraculous for distinct, randomly initialized systems to independently converge on the same high-dimensional structure unless that structure corresponded to the actual "joints" of reality. If model A (trained on Wikipedia) and model B (trained on Instagram) independently arrive at the same vector relationship between "King" and "Queen", the PRH assumes this vector exists in the world ($Z$), serving as the ground truth.

This stance represents the *mirror of nature* for artificial systems par excellence: the neural network is viewed as a mirror whose goal is to reflect the external world as accurately as possible. The "smudge" on the mirror (subjectivity, noise) is to be polished away by scale, leaving a perfect reflection of $Z$. It is precisely this assumption—that the convergence of the reflection proves the independence of the object—that our constructivist position will challenge.

For many proponents of convergent realism, the *territory* is a mind-independent reality to which our linguistic or mathematical terms successfully refer. To maintain this link across changing paradigms, realists like Putnam (1982) invoke a "principle of charity", arguing that even as our theories evolve, they continue to refer to the same stable entities (e. g., electrons or $Z$). The PRH adopts this charity by assuming that disparate AI models are all successfully "hooking onto" the same statistical invariants of the physical world.

It is worth noting, before we turn to our critique, that Huh et al. (2024) are considerably more careful than the framing metaphor of the cave suggests, and that any response owes them an engagement with these qualifications rather than a reconstruction of a cruder position. Their formal argument holds strictly only for *bijective* observation functions—the idealized case in which no information is lost in the passage from $Z$ to data—and they explicitly flag that real sensors are lossy and modality-specific. They further concede what they call "sociological bias in producing AI models": that researcher preferences and the collective priorities of a research community shape which architectures, objectives, and datasets get built and compared in the first place, and that convergence measured across such a population may reflect this shared history of design choices as much as it reflects any target those choices converge on. Both concessions are treated, however, as contingent limitations of present practice—sources of noise that more data, better sampling, or more diverse benchmarking could in principle reduce. Our argument in Section 5 is aimed precisely at this residual optimism: we will argue that the gap it concedes is not the kind of gap that scale can close.

## 3 Deconstructing the Mirror of Nature

It is crucial to observe that Putnam eventually distanced himself from robust scientific realism, entering a period of philosophical transition characterized by a search for a stable label for his "third way". This evolution saw him move from internal realism to pragmatic realism, and finally to natural realism—a position Case (2001) aptly terms *pluralistic realism*. Alongside his longtime interlocutor Richard Rorty, Putnam became a prominent critic of the very *no miracles argument* (NMA) he had once helped popularize, eventually conceding that the argument is likely circular.

The circularity of the NMA arises from its meta-theoretical structure: it begins with a second-order observation of the predictive success of science and employs *inference to the best explanation* (IBE)—or abduction—to conclude that the approximate truth of theories is the cause of this success. However, this reasoning presupposes that "being the best explanation" is a reliable guide to truth. By using abduction to validate the scientific method, the argument effectively relies on the very mode of inference it seeks to justify against the skeptic.

For over two decades, Putnam sought to develop a middle path between metaphysical realism and antirealism. This pluralistic realism maintains that while we have a plurality of "right" descriptions of the world, language is nonetheless capable of representing a reality external to thought. On one hand, Putnam respects realism's constitutive insight: that thought and language can represent parts of a world that are not themselves products of thought or language. On the other hand, he rejects the "God's eye view", arguing that there are multiple, irreducible, and equally valid descriptions of the same facts.

At the heart of this position is the doctrine of *conceptual relativity* (Putnam, 1987), which posits the *interpenetration* of fact and convention: a true statement cannot be cleanly split into a purely factual component and a purely conventional one, since the placement of a concept is itself determined by human interests. Putnam's stock illustration is mereological: a world of three discrete elements $(x, y, z)$ contains three objects for a standard mathematician, but seven for a logician who additionally counts every possible sum of parts as an object. Both descriptions are, for Putnam, "equally true"—there is no mind-independent "real" number of objects outside a chosen conceptual scheme, just as scientific essences such as water being $H_2O$ are products of our referential intentions, not discoveries independent of them. Truth, on this view, is a "joint product" of the world and the creative role of language users—which is already a considerably weaker position than the flat correspondence the PRH's rhetoric of "shadows" and "recovery" suggests.

Richard Rorty, however, would argue that Putnam's pluralistic realism is an unnecessary attempt to "cling to the ghost of objectivity". While Rorty shared Putnam's rejection of the "view from nowhere" (Rorty, 1979), he critiqued Putnam from a strictly anti-representationalist position. For Rorty, once the "ready-made world" is abandoned, we should cease worrying about whether language "represents" reality at all. He famously argued that "truth" is not a relationship between language and the world, but rather a "compliment" paid to sentences that our community finds useful (Rorty, 2016)—a move from truth to *solidarity*[1].

Where Putnam sees facts as constrained by the world, Rorty views "constraint" as a metaphysical leftover. To Rorty, the realist's appeal to constraint is a category error that conflates causal pressure with justification. He admits that the world exerts causal pressure on us—just as a floor constrains our physical movement—but denies that this pressure dictates the truth of our sentences. As Rorty famously put it:

> "The world does not speak. Only we do. The world can, once we have programmed ourselves with a language, cause us to hold beliefs. But it cannot propose a language for us to speak." – (Rorty, 1989).

Consequently, for Rorty, the distinction Putnam draws between "stars" and "tables" is merely an attempt to privilege scientific discourse over other cultural practices. By reframing knowledge as a biological adaptation, Rorty replaces the realist's *copying* with a Darwinian *coping*. In this neo-pragmatist view, we are not "beings who help define the world", but animals who have developed complex linguistic habits to help each other survive and thrive rather than accurately represent an intrinsic nature.

As a pragmatist, Rorty rejected proving his position, since for him this would itself be a categorical error; he offers pragmatic arguments instead. Realists, he argues, define the criteria for success in a way that only their own method can meet—you cannot use the internal logic of a vocabulary to justify that vocabulary over all others, any more than you could use the rules of chess to prove that chess is the only game worth playing. Choosing a way of speaking is not a logical decision but an intuitive, practical, or aesthetic one. This is why Rorty also rejects *foundationalism*, the architectural metaphor of knowledge as a building resting on self-evident truths, drawing on Quinean holism's rejection of the analytic-synthetic distinction (Quine, 1951): in a holistic web, no single strand is an absolute foundation, and asking for a "rigorous proof" of the whole web

[1] Seen as desire for as much intersubjective agreement as possible.

is asking for a standpoint outside it—a "God's-eye view" Rorty insists is unavailable. There is no "outside" to our linguistic practices, only the ongoing conversation of a culture; changing conceptual schemes is redescription, not the discovery of a firmer foundation. Rorty's rejoinder to the charge that this is mere relativism is that it is instead *ethnocentrism*: we speak from within a tradition, and our "rigor" consists in the conversational habits that tradition has found useful.

In summary, while Putnam attempts to repair the *mirror of nature* so that it might better reflect a pluralistic reality, Rorty insists on shattering the mirror entirely. He urges us to stop worrying about "rigor" as an accurate mapping of the world and to redefine it as the pursuit of human flourishing.

It is worth being precise about what this section has and has not shown. It is not, primarily, a claim that the PRH is philosophically naive because better-informed readers already know realism is complicated; sophistication is not by itself an objection. The point is narrower and more useful: even within the tradition that most wants to preserve a notion of objective truth, the flat correspondence picture the PRH's language of "recovery" and "shadows" trades on had already been abandoned as untenable by its own strongest defenders, decades before the PRH restated it for neural networks. This licenses treating the PRH's implicit epistemology as a regression rather than a live option in contemporary philosophy of science, which is a different and stronger claim than merely finding it unfashionable. It is also worth distinguishing this line of criticism from a more familiar one in science and technology studies, where AI's failure to represent the world neutrally is typically diagnosed as a matter of *social bias*: training corpora encode the perspectives, omissions, and power relations of those who produced them, and this can in principle be documented, contested, and corrected (cf. Bender et al., 2021). That critique, and the actor-network tradition's insistence on tracing what gets black-boxed inside a technical artifact (Latour, 1987), share our suspicion of the PRH's realism, but they locate the problem at the level of *content*—which data, whose perspective, what gets omitted—and hence treat it as contingent and correctable. Our argument, developed via Luhmann in the next two sections, is not about which communication gets fed to a model, but about the structural relation between any communication whatsoever and the reality it is about. That relation does not improve with better data; it is what operational closure means.

## 4 Operational Constructivism

While Putnam attempted to balance mind and world, and Rorty dissolved the world into a social conversation, they both stick to a humanistic conception of society made of people, that is, linguistic agents who reason, suffer, and converse. Both moved past old-school ideas of universal truth and the *mirror of nature* but remain firmly anchored in a humanistic framework. To go even further and beyond the subject/object dichotomy, we now turn to the *operational constructivism* of Niklas Luhmann and his theory of self-referential autopoietic systems. This section introduces the apparatus we will need in Section 5: *distinction* and *observation* give us a vocabulary that does not presuppose a subject; *operational closure* and *structural coupling* let us say precisely what a model does and does not have access to; the treatment of *reality as recursive stabilization* lets us say what convergence actually stabilizes; and the account of science as a *truth-coded* system, in the final subsection, lets us dissolve the no-miracles argument itself rather than merely oppose it.

Before we present our observation of his theory, we should recognize that it is not immediately obvious why a sociological theory would tackle questions of epistemology. After all, Luhmann was not an official philosopher and never regarded himself as one. However, he was awarded one of the most prestigious philosophy prizes in Germany, the Hegel Award, in 1989, and in his works he referred to Plato and Kant at least as often as to the works of Weber and Durkheim (Möller, 2011). He argued that a theory of society cannot blindly adopt the established scientific principles of the very society it seeks to explain (Luhmann, 1998). Consequently, it has to ask questions that were usually reserved for the philosophy department. As we will see, Luhmann provides an update to the Old European tradition; an update that we deem particularly potent for understanding data-driven artificial intelligence.

Inspired by Spencer-Brown (1969), Luhmann starts with the act of *observation*: he draws a distinction (system/environment) and indicates one side (the system). His beginning cannot be proven to be "correct" because any such proof would presuppose a theory of certain complexity (Luhmann, 1988). Thus, his beginning seems explicitly axiomatic and self-consciously "naive":

> "We assume that cognitive systems are real in a real environment, meaning that they exist. Many say such an assumption is naive. But how shall we begin if not naive? A reflection of the beginning can not be at the beginning but can only be executed within a theory that has reached a certain complexity." – Luhmann (1988)

In accordance to Spencer-Brown (1969), he is rather instructive or shall we even say *operational*. We can follow along and may recognize that what we construct (a theory of observers) presupposes an observer who does the construction. Consequently Luhmann's approach is inherently *autological*: it recognizes that a theory of society (and cognition) cannot adopt the established principles of the society (and cognition) it seeks to explain. Instead, the theory must account for its own existence as a scientific enterprise within the social space it describes (Luhmann, 1998).

Following Spencer Brown's calculus of form, cognition is viewed not as a bridge between knower and known, but as a *form of distinction* (Spencer-Brown, 1969) that, according to Luhmann, can happen without spirit or mind. By drawing a distinction and indicating one side of it, both the observer and the observed emerge simultaneously.

If one now assumes that the given world guides observations, then two different observers will observe the same thing; in that case, it is not particularly interesting to observe their observations (Esposito, 1993). However, a world with multiple observers can not be identical for anyone because for each observer all other observers belong to their respective environment. Furthermore, multiple observers open the path to second-order observation, i. e. observers can observe each other.

Systems theory does not start with a world but with an act that one could argue is prior to it, that is, *observation*: observation constitutes its subject-matter. In other words, Luhmann starts with constructivism which provides the epistemological analysis of the mechanisms that are capable to coming up with observations that constitute reality (ontology). In this sense (and only this sense), epistemology comes before ontology. Consequently, the question is no longer whether knowledge *matches* an object, but how both sides *emerge* through the same operation and how society works if we drop the glue of intersubjectivity.

Luhmann then integrates insights from *radical contructivism* (Glasersfeld, 2002) into his theory of autopoietic self-referential and operationally closed systems, arguing that cognition is only possible *because* it has no operational access to reality. He wrestled with the same question Kant (1781) asked a few hundred years ago:

> "How can cognition determine objects outside itself? Or: How can cognition determine that there is something independently from itself existing, if everything that cognition determines already presumes cognitive capacity and cannot be identified independent from cognition through cognition?"
> – Luhmann (1988)

Kant's transcendental idealism answered the question by centring everything around the subject—the human mind and its internal structures. In reaction to Hume's scepticism Kant claimed to have found some structures that are *a priori* as well as *universal*. While we have to be cautious about the phenomena we recognize, Kant assures us that objectivity can be saved by the structures we humans all must share, which are the conditions of the possibility of experiencing anything at all.

Luhmann gives a different answer: openness via closure. He keeps Kant's idea that the mind actively structures the phenomena but dethrones the transcendental subject, Hegel's absolute spirit as well as universal reason and replaces it with *observing systems*. Rationality, Luhmann claims, is always the contingent product or construction of a system and that there is no such thing as *universal reason*.[2] A system has no *operational access* to what it is not, that is, society has no access to something that is not social, e. g. reason, and the mind has no access to something that is not mental, e. g. communication. Consequently, there are no universals that apply to all systems, except the process of distinguishing themselves.

In a very Kantian move, Luhmann asked about the conditions of the possibility that systems exist. Starting with distinction, he reasons that a system has to be *operationally closed* because it is the only way to be *cognitively open* (and we can observe cognitively open systems in our environment). If a system were not closed, it could not process the complexity of its environment. It would not exist and we could not have observed it because it could not continue its autopoiesis. In other words: Operational

[2]Similar to the pragmatists

closure prevents the system's environment from overwhelming its internal processes. For example, the membrane does not allow for the environment to directly take part in the cell's biological operations. And a mind does not allow for the environment to directly take part in thinking. If we want to understand how the brain works, we have to describe physiological and biological processes, not thoughts and social phenomena, which, while they may influence these processes as environmental factors, remain operationally distinct from them (Möller, 2011, p. 22). It is not helpful scientifically to say that human brainwaves are primarily human rather than cerebral or that the computation of artificial neural networks are mental rather than computational. We will return to this point directly in Section 5.2: whatever a large model's loss function measures, it does not measure a comparison between the model's internal states and physical reality, for exactly the reason a brain does not compare thoughts to atoms.

Instead of a static mind, organism or social system, Luhmann describes systems that constantly produce themselves through their own operations. We observe a system as the distinction (an operation) between system and environment—it is the act of drawing a line in the world to separate one side from the other. It is observed as a *form* (Spencer-Brown, 1969)—a unity of the distinction. The system's operations can only connect to each other and to nothing outside of itself, that is: systems are *functional differentiated*. An organism consists of what it ate and a mind operates with what it had thought before (Luhmann, 1993). A mind cannot continue a thought with a brainwave and communication can only continue with more communication. Systems do not "exist" like a rock; they happen through continuous operations. Their *structure* constantly change while their *organisation*, that is, their autopoiesis keeps going until they vanish.

Continuous decay becomes an indispensable contributing factor to the system's persistence. If, for example, every thought were to remain stationary in the consciousness, the system's capacity for order would be overwhelmed within minutes (Luhmann, 1995). A thought or a communication is an event that vanishes at the very moment of its appearance. A thought cannot be "held onto" without it ceasing to be a living process. But when the elements of a system (thoughts) decay immediately, a vacuum is created. This vacuum generates the necessary pressure for connectability (Anschlussfähigkeit): because the current thought fades, the system must produce a subsequent thought to avoid ceasing to exist. Thus, continuous decay is not the end of the system, but the fuel that forces it to work constantly. Selection is required: something is chosen, and other things are omitted. For the system to select something new, the old must vacate its space. A system is therefore not stable because it conserves things, but because it organizes constant change. And since events cannot be repeated, expectational structures—such as language—are required so that the system does not have to start from scratch every time.

Relying on the concept of distinction instead of "objects out there" systems theory eloquently dissolve the mind-body problem (Möller, 2011) as well as the split between idea and matter by increasing the clarity of the terminology. Of course, "dis-solution" can only mean: an observational construct. Because any distinction has a boundary and any boundary has two sides, distinction is perfect continence (Spencer-Brown, 1969).

In other words, you cannot have a system without an environment, thoughts without content, a signifier without a signified, or an environment without a system. By isolating both as a unity we created a problem that can be resolved by rejecting the premiss of the question and asking a different one. Thus, Luhmann also implicitly critiques radical constructivists of his time who tend to neglect one side of the distinction[3]—effectively rescuing constructivism from the danger of solipsism by constructing an anti-humanist theory that is so radical that it paradoxically becomes a form of realism (Möller, 2011).

In this "operational twist", Luhmann abstracts *autopoiesis* from its biological origins (Maturana & Varela, 1980). Whereas Maturana anchors the observer in self-producing physio-biological organisms, Luhmann defines systems as sequences of transient *events*. For Luhmann, cognition no longer presupposes a mind (Möller, 2011, p. 39) but can be realized by living, psychic (mind), social (and maybe artificial[4] or technical[5]) systems. These abstractions allowed Luhmann to use biological concepts for a theory of society—and others (Reichel, 2011; Watson & Romic, 2024) to conceptualize technology as an autopoietic system.

This shift necessitates a deconstruction of the "human being". Again, Luhmann strictly differentiates between *living systems* (the human body), *psychic systems* (thoughts, feelings, etc.) from *social systems* (communication).

> "One can not presuppose that psychic systems or social systems are living systems. This does not follow from the fact that psychic and social systems presuppose life." – Luhmann (1995)

Consequently, his theory claims that humans do not communicate; only communication can communicate.[6] It is his answer for the conditions of the possibility of society: not intersubjectivity and a shared understanding based on reason but the co-evolution of communication and mind (and probably machines) that constantly irritate themselves and misunderstand each other. The individual's inner life—thoughts and feelings—remains operationally closed; it cannot be "instructed" by the social world. Instead, psychic and social systems *irritate* one another across their boundaries.

Crucially, operational closure is not causal isolation but it differs from the traditional input-output model. Systems are *structurally coupled* to their environment—the

[3]One can not cancel out the object within the subject/object distinction, reality from the idea/reality distinction, or the signified from the signified/signifier distinction without collapsing the whole distinction.

[4]If we define "artificial" as "something that is designed" then the term "artificial system" is an oxymoron. However, did "we" design the computations which happen? One of the questions in artificial intelligence then is if autopoiesis can happen and can lead to such a complexity in structure that something like a minds emerge even if its substrate—to which it is structurally coupled—is designed.

[5]Luhmann speculated (without elaboration) about the emergence of novel technological autopoietic systems that may operate on the base of computer technology, see e. g. (Luhmann, 1998, p. 117).

[6]The fact that humans do not communicate, opens the path for what Esposito (2022) calls "artificial communication", that is, the possibility that machines be structurally coupled to communication in such a way that they irritate it.

concept that will let us say, in Section 5.3, what a foundation model's environment actually consists of, and why "more data" changes what it is coupled to without changing the kind of access closure permits. For example, minds and communications are coupled via language and operate in the medium of meaning (Sinn). While a system cannot register external causality directly, it translates environmental *perturbations* (causal kicks) into internal *irritations* (meaningful noise). Information, therefore, is not a substance that flows from environment to system, but an internal event—a "difference that makes a difference" (Bateson, 1972). This does not invalidate Shannon's mathematical theory of communication (Shannon, 1948). However, Shannon does not describe meaning-making, but rather the physical substrate and reliability of the medium through which these irritations are processed. Computers transfer information between each other because they "irritate" one another in a predetermined way. Together, physical instrument, codes and protocols usually define how this is actualized and sometimes fails, which causes surprise for the observer—the functional simplification reveals itself. However, autopoietic systems are neither designed nor can we assume that they evolve in such a way that perturbations cause the same irritations. In fact, we observe the opposite. For example, bodies, minds, the law, the economy, the media, and science arguable process the perturbations caused by climate change very differently. And Society's attempt to trivialize the structural coupling between body, mind, and person—the entity we label "human being"—inevitable encounters the recalcitrance of the individual.

What then *is* reality in a theory of autopoiesis and operational closure? The short answer: reality is an effect of observation (but observation is not substantially defined) (Möller, 2011, p. 79). It is the recursive stabilization of a system's internal operations against the "noise" of its environment. It is the result of a system failing to contradict itself.

Let us elaborate: Since Kant (1781), reality has been understood as resistance—as something that challenges the mind and thereby demands explanation. However, operational closure seems to make such challenges impossible. Luhmann argues that we should not abolish Kant's idea of resistance but we should move it into the system because—with the exception of the system's destruction—only operations can resist operations. Thus, resistance is not imposed by an external world but relocated within the system itself. Resistance, which is experienced as information, is the result of resolving an internal conflict—the result of the system's operations resisting the operations of the same system (Luhmann, 1998). By this relocation, we do not lose the world. It exists but it is *unmarked* (Spencer-Brown, 1969) i. e. a horizon (Husserl, 1939, 1954) of pure complexity. It is the unobservable unity that requires a distinction to see something that is already no longer "the world". An observer cannot stand "outside" it to see it.

This assertion, that reality is the recursive stabilization of a system's internal operations, aligns with the mathematical logic of recursive functions. When a system performs operations on its own previous operations, it functions as a recursive loop

$$x_{t+1} = f(x_t).$$

In this formula, $f$ represents the system's operation and $x$ represents the system's state. Stability—and thus reality—is achieved when the system reaches an Eigenvalue. This occurs when the operation no longer changes the state, expressed as

$$x = f(x).$$

At this point, the system "fails to contradict itself". The resulting Eigenvalue is what the system then treats as a persistent "object" or a "fact" (Heylighen & Joslyn, 2003; von Foerster, 1974). Consequently, what we call "objects" are not things existing independently in the environment, but are tokens for the consistency of the system's own recursive operations. This is the sense in which we will later read the PRH's "platonic" latent space—and, concretely, its color representations in Section 5.4—as an Eigenvalue of a training process rather than a discovery.

Something is real for a system if the respective pattern of structure can be replicated. While the individual operation decays immediately, the replicability of structures (such as language or scientific methods) ensures the stability of the system's reality. Reality is thus that which has proven itself within the system as a repeatable pattern.

This recursive operation is finalized through what Spencer-Brown (1969) calls *re-entry*. Re-entry occurs when the distinction between system and environment is copied back into the system. This allows the system to use its own distinction to distinguish between itself (self-reference) and its environment (other-reference) within its own operations. Through re-entry, the system can treat the environment as real because it observes the environment as the other side of its own internal boundary. The system can compare its internal state with its observation of the environment. The friction between these two (the "noise") is what generates information.

Furthermore, because a system can only see what its specific distinctions allow it to see, it cannot see what it cannot see. There is always a *blind spot* that constitutes the system's unique reality. This blind spot is not a deficit to be overcome, but the constitutive requirement for observation. To see anything at all, the system must use a distinction; however, it cannot simultaneously see the distinction it is using. At the level of first-order observation, reality is the experience of resistance. It is the "What" that is encountered when a system's internal operations run into a contradiction. For example, when our minds felt a contradiction within metaphysical realism.

As second-order observers (of systems), reality is the world-certainty that arises because the distinction being used to observe remains latent during the act of observation. Everything a system observes is its construction, yet most of the time it has to be observed as if it were not a construction such that it can continue its operations and not get stuck in a hyper-reflective loop.

Consequently, the theory breaks with the notion of a common reality that is somehow "represented" within all systems or elements that take part in reality. Operational closure means that systems construct themselves and their own realities (autopoietically). It enables systems to generate an openness towards their environment, but no operational openness. It allows systems to build up mechanisms with which they can observe their environment.

How an eye sees the environment, what a brain makes of the visual irritation it receives, is entirely dependent on its own structure. It reduces complexity of what can be seen in a specific way and thus builds up internal systemic complexity. Each eye, in accordance with its own internal structure, constructs a unique environment. Two humans don't see the same tree because the tree "enters" their minds. Instead, they both have similar internal machinery that reacts to light in a similar way and they share a language such that communication overwrites differences—they share the same evolutionary history of "not being destroyed" by their environment.

Questions about who sees the world more adequately or correctly become obsolete because there is no common reality for comparison. How a system is real depends on its own self-production, and how it perceives the reality of its environment also depends on its self-production. Furthermore, by distinguishing itself as a system, a system also constructs its understanding of its environment. And thus a systemic world cannot suppose any singular, common environment for all systems. By starting with distinctions instead of objects, reality becomes a multitude of system-environment constructions that are in each case unique. This is the radical shift away from traditional Old European epistemologies and ontologies.

The operative functionality of science, in this view, is not to reach a terminal state of total knowledge, but to refine the system's capacity for handling complexity. Paradoxically, science produces a *surplus of ignorance*: every time a theory identifies its blind spot—the specific distinction it uses to observe—it must introduce a new distinction to account for that blind spot, thereby expanding the realty it constructs rather than reducing it to a singular "Truth".

To refine this, we must recognize that science as a social system operates via the binary code of truth/untruth. Science does not discover truth in the environment; rather, "Truth" is the internal program the system uses to select which communications can be connected to the next. This code allows the system to achieve operational closure: only scientific communication can determine what is scientifically true. Even pseudo-scientific arguments work on the truth/untruth distinction and not, for example, on the ugly/beautiful distinction. If creationists want to prove their point, the communication that happens, happens within science. But if they use the transcendence/immanence distinction, the communication is religious.

Science distinguishes itself from other social systems (like law or politics) through a specific re-entry: it brings the distinction between "theory" and "method" back into its own operations. Re-entry occurs when science observes its own observations. By recognizing its own blind spot (but only scientifically), science turns its constructive process into a theme of investigation. For Maturana (2000) science is operationally closed because it deals only with the explanations of experiences which are themselves experienced:

> "Science is constitutively a domain of explanations that requires no assumptions of an external reality, because it deals only with the coherences of experiences." – Maturana (2000)

Since, for Luhmann, science does not consists of experience because it does not consists of humans, he updates Maturana's viewpoint shifting it towards communication. Of course, experience (first-order observations) and the explanations of experience (second-order observations that hopefully accomplish reproducibility) causally influence science but do not steer it, because experience is not communicating. A scientific fact is not a reflection of an object, but a stable point of recursive communication that has survived repeated attempts at contradiction. The communication might be about an "object" like "electrons" which emerge in communication due to observations achieved by certain instruments that irritate the scientific system. Science observes "electrons" as the recursive stabilization of its own measurements. However, the environment (the physical reality) cannot "tell" the scientist that electrons exist because the system is operationally closed. As Rorty proclaimed: "The world can not talk." Therefore, a theory is real not because it matches an external world, but because it has reached a state of recursive stability where the system can continue its autopoiesis without internal collapse.

The high status of physics in the hierarchy of sciences can be explained through this dual-aspect reality. At the first-order level, physics encounters a high degree of resistance through structural coupling with sophisticated measurement apparatus. This "stubbornness" of the experimental result is what the physicist calls "reality". However, at the second-order level, we observe that the "Laws of Nature" are actually Eigenvalues—stable, recursive points of communication that have achieved a state of non-contradiction through mathematical formalization. The certainty of physical laws is not a reflection of a latent world, but a testament to the system's ability to keep its own constructive mathematical distinctions latent during the act of measurement.

The success of physics is not a result of it having finally grasped the unmarked world. Rather, it is the result of science reaching a state of maximum operational consistency. Through the use of mathematics and experimental technology, physics has refined its internal Eigenvalues to such a degree that the resistance of the environment is channeled into a binary code of truth/untruth with near-zero ambiguity. Physics works not because it represents the territory, but because its map has become so structurally dense that it can no longer be contradicted by the irritations it receives.

Consequently, any "is" statement written down in our work is communication within the scientific system. These statements might, and hopefully will, connect to further communication. We are observing a theory and how different systems are being observed through the glasses of this theory, offering the possibility to observe our observations. Our work is not "revealing" the truth about reality; it is constructing a scientific description of how descriptions are constructed. It is scientifically real if the scientific system fails to resist it.

### 4.1 Dissolving the Miracle

We can now redeem the promise made in Section 3. Putnam and Rorty came to treat the no-miracles argument as circular: it observes the predictive success of science and infers, by abduction, that success is best explained by approximate truth—but this already presupposes that inference to the best explanation is a reliable guide to truth, which is the very thing at issue. Diagnosing a circularity, however, still leaves the intuition standing: it still *feels* miraculous that our theories work so well if they are not tracking something real, and Putnam and Rorty each spent a career proposing something to put in its place, natural realism and solidarity respectively. Operational constructivism can instead dissolve the intuition rather than replace it.

The dissolution follows directly from treating science as a truth-coded social system, as we have just done. The no-miracles argument takes a second-order fact—that science, observing its own history, describes its most stable and widely connected products as "true"—and reads it as evidence of a first-order relation between theory and an independent world. But a system that operates on the truth/untruth code will, definitionally, mark its most successful products "true" and use exactly that marking to license further connection: further citation, further application, further theorizing. That its most successful theories come out marked true is not a surprising discovery about science's relationship to reality that then needs an explanation; it is a restatement of what the code does. There is no further fact—no correspondence, no approximate truth, no miracle averted—left over once the tautology is seen for what it is. This is a stronger position than Rorty's: Rorty replaces the demand for truth with the observation that our vocabularies are useful to us, which still leaves the appearance of a miracle needing a redescription (usefulness rather than truth). Systems theory denies there was ever a miracle to redescribe.

The PRH restages exactly this move for machine learning, and exactly the same dissolution applies. Training and evaluation are themselves a code—a loss function or benchmark score that marks certain internal configurations as more "successful" than others and licenses further gradient steps, further citation, further deployment on that basis. That models selected under a shared code, and increasingly trained and evaluated against the same benchmarks and the same corpora, come to resemble one another and to satisfy that code's own criteria of success with growing consistency is not a fact that calls out for an explanation in terms of a shared, mind-independent $Z$; it is what selection under a shared code produces, whatever it is a code for. Section 5 develops what convergence *is* evidence of, once this deflationary move has cleared away what it is not evidence of.

## 5 System-Theoretical Perspective on the Platonic Representation Hypothesis

Based on the theory of operational constructivism, we can now provide a systematic view of the platonic representation hypothesis (PRH). By re-evaluating the concepts of closure, coupling, and viability, we can demonstrate why the isomorphic convergence of models does not necessitate the convergence towards *the* "really real reality".

## 5.1 Artificial Systems?

Before we can put operational closure and structural coupling to work against the PRH, we must confront a problem that the rest of this section would otherwise inherit silently: is it even legitimate to describe a large model as "operationally closed" in Luhmann's sense at all? Operational closure, as introduced in Section 4, is a predicate of *autopoietic* systems—systems that produce the very elements out of which they consist through their own operations, and that achieve self-reference only through re-entry, the folding of the system/environment distinction back into the system's own operations. If large models are not autopoietic systems in this strict sense, then talk of their "closure" risks being a metaphor doing more work than it has earned. We take this problem seriously rather than deferring it, because its resolution turns out to sharpen our argument against the PRH rather than to weaken it.

The classification of technology as "system" in Luhmann's sense remains a genuine point of debate (Baecker, 2025; Lovasz, 2023; Reichel, 2011; Watson & Romic, 2024; Zönnchen et al., 2025). Maturana and Varela (1980), from whom Luhmann borrows the concept, would certainly deny artificial neural networks the attribute "autopoietic": although they self-organize, their cognitive performance is a result of self-organization occurring during a training phase *within* a pre-existing system built with specific, externally imposed features (Bianchini, 2023)—the model does not produce its own architecture, training procedure, or objective function. If we look more broadly toward technology in general, including the production of machines by machines and software by software, the problem of attribution becomes more difficult: Reichel (2011) and Lovasz (2023) frame technology itself as a sovereign, self-referencing system operating on the binary code *work/fail*, and Watson and Romic (2024) describe large language models as a "subsystem of the technology [...] which in modern society mediates between individual thinking, the physical world, and between thought and society."

We follow Baecker (2025) in resolving this in the negative. Under the label "computer", Baecker (2025) classifies technological artefacts such as large language models as *trivial machines* in Heinz von Foerster's precise technical sense (von Foerster, 1984). A trivial machine is synthetically determined, analytically determinable, and—crucially—history-independent: once its transfer function is fixed, the same input under the same conditions reliably yields the same output, and its past operations leave no trace that changes how it will respond next. A *non*-trivial machine, by contrast, is history-dependent: its internal state changes as a function of its own prior operations, so that the same input can yield different outputs depending on the machine's history—which is exactly what re-entry produces in an autopoietic system, and exactly what makes such a system's own future observably underdetermined by its environment. On this diagnosis, large models lack *reflective self-referentiality* specifically because they lack re-entry, and hence the distinction between *self-reference* and *other-reference* that re-entry alone makes available to a system (Zönnchen et al., 2025).

This classification is not merely compatible with the PRH's central empirical finding; it is better evidence for it than the PRH's own explanation. Consider what convergence would look like under each hypothesis. If large models were non-trivial, autopoietic systems—individuating their internal structure through their own operational history, the way a mind or a society does—we should *expect* idiosyncrasy, not convergence: two systems exposed to overlapping but non-identical histories would drift apart precisely because their present state depends on a self-referential trajectory that the environment underdetermines. Convergence toward a single statistical fixed point across differently initialized, differently architected models trained on non-identical corpora is, on the other hand, exactly what a population of trivial machines does when exposed to a shared code and a shared statistical environment: their outputs are fully determined by input and architecture, so sufficiently similar inputs under a sufficiently similar training regime reliably drive sufficiently similar transfer functions toward the same fixed point. The PRH reads convergence as a system escaping its triviality by discovering something outside itself; read through von Foerster, convergence is closer to what we described in Section 4 as a system "failing to contradict itself"—the signature of triviality, not its refutation. Where the empirical results in Huh et al. (2024) strengthen the argument that large models are not autopoietic, it is *because* they converge toward the same statistical model: if this turns out to be a general pattern, these systems are either trivial to begin with, or they get *trivialized by scale*, since further training only tightens an already input-determined transfer function around its fixed point.

This leaves a genuine and important asymmetry with the rest of our vocabulary, which we do not want to paper over. *Closure* in the full Luhmannian sense—the recursive, self-referential production of a system's own elements—does not apply to a trivial machine, because there is no self that the machine's operations reproduce. What survives, and what our argument in Sections 5.2 and 5.3 actually needs, is a weaker and more defensible claim: that a trivial machine's output is never a direct read-out of its environment but is always relative to a fixed, internally given transfer function through which every input must first pass. An environment can perturb a trivial machine—change what goes in—but it cannot dictate the function that maps input to output; that function is given once, at construction, and is precisely what training fixes. This is enough to block the PRH's inference without requiring the stronger claim that models are autopoietic subjects in their own right, and it is compatible with Esposito (2022)'s account of *artificial communication*: algorithms need not be sense-making, self-referential systems to participate in communication, because communication does not require a communicating subject on either end—only a structurally coupled artefact whose outputs psychic and social systems can take up, interpret, and connect to further communication. On this view, large models occupy neither the position of mere passive tools nor that of autopoietic, sense-making agents; they are trivial machines that participate in communication precisely by being reliably, unremarkably input-determined, which is what makes them useful conduits for redistributing the redundancy already sedimented in the communication they were trained on, rather than sources of meaning in their own right. This is the double claim the rest of this section develops: convergence is neither evidence that models are approaching a mind-independent reality, as the PRH holds, nor evidence that they are autonomously constructing a world of their own, as a careless reading of constructivism might suggest—it is evidence of trivial machines becoming maximally well-adapted conduits for an environment that was never physics to begin with.

### 5.2 The Epistemology of Closure

The PRH posits a unidirectional flow of information: reality ($Z$) generates data ($X, Y$), which the model ($M$) ingests to reconstruct $Z$. This model assumes that $Z$ is accessible, provided the observer is sufficiently perceptive. Constructivism, however, fundamentally rejects this possibility.

Under the framework of *operational constructivism*, cognition presupposes a system-environment boundary, implying that meaning is strictly an internal construction. No cognitive system—whether biological, social, or artificial—possesses direct access to the world-in-itself ("das Ding an sich" Kant (1781)); it accesses only its structurally coupled environment, which is always already a selection performed by the system—an order generated from noise.

For neural networks, this constraint is absolute: they do not perceive the physical world ($Z$) but process the communicative distinctions (data) already mediated by society. The environment of a large language model is not the physical universe of atoms and forces; it is the symbolic universe of tokens and pixels. Consequently, the system remains operationally closed. During training, the model does not compare its output to the "really real reality," but checks internal states against other internal states—specifically, its own predictions against the "ground truth" labels provided by the dataset. The loss function, therefore, does not measure a correspondence to an external truth but a system-internal consistency within its data-environment. The system never "breaks out" of this recursive circle to verify whether its representation matches an external ontology; it merely optimizes for viability within the constraints of its training objective.

Recall from Section 2 that Huh et al. (2024) already concede something adjacent to this: that observation functions are lossy rather than strictly bijective, and that the population of models compared for convergence is shaped by "sociological bias in producing AI models". Both concessions treat the gap between representation and $Z$ as a matter of degree—a residue of noise, sampling, or shared design fashion that additional data, more diverse benchmarks, or a more representative population of researchers could in principle shrink. The argument from operational closure is not a more cautious version of this same claim; it denies the premise that makes "degree" the right category at all. A lossy or biased sample is still, at least in principle, a sample *of* $Z$, related to it by a channel that scale and diversity can widen. A structurally closed system, however, has no channel to widen: what it is coupled to is, by construction, always already a selection performed on the other side of a system/environment distinction, and no accumulation of further selections converts a sequence of selections into an unselected given. Feeding a model more data does not thin the veil between model and $Z$; it enlarges the corpus of stabilized communication the model is coupled to, which is a different reality with a different history, not a clearer view of the same one. This is why we describe the gap as constitutive rather than contingent: it is not that today's models see $Z$ dimly and tomorrow's might see it clearly, but that "seeing $Z$" is not among the operations a structurally closed system can perform, at any scale.

### 5.3 Structural Couplings of Large Models

Because of operational closure, learning can only happen through co-evolution as structural drift (Maturana & Varela, 1980). In the case of foundation models, the environment is the aggregate of documented communication. When Huh et al. (2024) observe that models converge, they are observing a system becoming tightly structurally coupled to the stabilized semantics of the social system. The "reality" that informs the loss function is not the laws of physics, but the recursive patterns of communicative selections.

From a system-theoretical perspective, the PRH appears to equate the internal stabilization of systemic observations with an objective external ontology. Within the Luhmannian framework, however, the concept of an "absolute" $Z$ is seen not as a discoverable entity but as an unobservable horizon. Consequently, the convergence of models might not represent an escape from the "cave" toward objective reality, but rather an increasing attunement to the recursive redundancies of social communication. Rather than uncovering an objective "truth", these models identify statistical *eigen-values* (von Foerster, 1974) that have emerged from the history of communication. Following Baecker (2025), this process can be understood as the computer's operation within the distinction of *predictive data* and *data space*. From this viewpoint, the "Platonic" latent space serves not as a map of reality, but as a stabilized navigation through the vast *data space* of communicative possibilities.

To clarify the nature of this convergence, we must differentiate between environmental complexity, understood as the "unmarked state" (Spencer-Brown, 1969) of raw resistance that remains operationally inaccessible to any system, and stabilized semantics, which comprise the reality of science, language, and logic. These semantics do not represent the "world-in-itself" but are historically evolved social constructions used by psychic and social systems to reduce environmental complexity into manageable meaning. Because an artificial neural network trains exclusively on these stabilized semantics rather than the unmarked state, its convergence is not a movement toward an objective Platonic reality. Instead, it represents an isomorphism between the model's internal latent map and the social map of communication, suggesting that the PRH describes a process where the machine becomes perfectly attuned to the redundancies of our own social constructions.

Large language models and other foundational models are cultural and social technologies (Bender et al., 2021; Farrell et al., 2025; Klein et al., 2025; Sorensen et al., 2024) that facilitate second-order observation (Esposito, 2022; Zönnchen et al., 2025). Rather than described as intelligent agents or autopoietic systems, these models can be more accurately understood as technical artefacts that *interpenetrate* with psychic and social systems (Baecker, 2025). In this capacity, they perform cognitive functions—specifically the high-level reduction of complexity—without possessing autopoiesis or reflective self-referentiality themselves (Zönnchen et al., 2025). They operate as performative participants in *artificial communication* (Esposito, 2022), facilitating second-order observation by providing stabilized patterns from a vast data space, yet they remain fundamentally anchored in their status as non-autopoietic, computational entities.

From this perspective, the PRH could be interpreted as reifying the structural stability of the map by treating it as evidence for a singular, objective territory ($Z$). In this view, $Z$ appears not as a Platonic form, but as an *operational construct*—an emergent statistical regularity produced by the massive redundancy of communicative selections. What the PRH perceives as a discovery of an external ontology may instead be seen as the mathematical crystallization of the social system's own recursive descriptions.

### 5.4 A Case in Color: Rereading the CIELAB Experiment

The argument so far has been conceptual. Huh et al. (2024) also offer their most concrete piece of evidence for a physically grounded $Z$, and it is worth rereading directly rather than leaving the constructivist argument at the level of general principle. Building on Abdou et al. (2021), they compute three independent representations of color and show that all three recover essentially the same layout: (i) the perceptual color space CIELAB, engineered to be perceptually uniform, so that equal distances in the space correspond to equal perceived differences in color; (ii) a representation derived purely from how often pairs of pixel colors co-occur near one another in natural images (CIFAR-10); and (iii) a representation derived purely from how often color words co-occur near one another in text, using both contrastive (SimCSE) and predictive (RoBERTa) language models. That cooccurrence statistics computed independently over pixels and over words recover, via multidimensional scaling, essentially the same structure as CIELAB is offered as their cleanest demonstration that vision and language are triangulating one and the same underlying reality—in this case, quite literally the physics of color.

The Luhmannian rereading does not dispute the finding; it disputes what the finding is a finding *about*. CIELAB is not a direct transcription of the physical continuum of light. It is a standardized, historically specific model of human color *perception*, developed by the Commission Internationale de l'Éclairage from psychophysical experiments in which human observers judged which color differences looked equal to them, and explicitly engineered for perceptual uniformity relative to human discrimination thresholds rather than for uniformity relative to wavelength or spectral power. It is, in our vocabulary, a stabilized structural coupling between the psychic systems of human color perception and the scientific-cum-standards communication that measured, contested, and eventually ratified it—a scientific Eigenvalue of exactly the kind described in Section 4, not an unmediated readout of the "unmarked" physical world.

Once this is seen, the convergence of the other two representations onto CIELAB stops looking like two independent triangulations of physics and starts looking overdetermined by a shared history that was never physical in the first place. The pixel-cooccurrence representation is not built from raw photon counts; it is built from CIFAR-

10 pixels, which are the output of a camera sensor, a demosaicing algorithm, and a color-correction pipeline, all of them engineered—like CIELAB itself—to match human perceptual expectations about color rather than to preserve physical spectral information. The word-cooccurrence representation, meanwhile, is built from a vocabulary that Huh et al. (2024) themselves source, via Lindsey and Brown (2014), from English speakers naming color chips: a lexicon that is, by construction, organized by exactly the perceptual apparatus CIELAB was standardized to model. None of the three representations in this experiment is closer to $Z$ than the others in the sense the PRH needs; all three are technical traces of the same underlying coupling—human color perception and its scientific standardization—encountered through three different channels (a color-difference survey, a camera pipeline, a naming survey). That cooccurrence statistics computed over any two of these channels recover the third is unsurprising once the channels are seen for what they are, and it requires no inference to a shared physical reality behind all three to explain it.

This is, in miniature, exactly the trivial-machine result of Section 5.1: a photographic dataset and a color-naming corpus are two data-environments already saturated with the same human-perceptual and scientific-communicative history, so two systems trained on cooccurrence within each should be expected to arrive at highly similar fixed points, without either system needing any access to color beyond what has already been metabolized, standardized, and sedimented by human visual and linguistic communication. The PRH's own strongest case study is, on inspection, a case study in structural coupling to a shared communicative history—not in the discovery of physics.

### 5.5 Viability over Validity

While the PRH suggests that convergence implies truth (validity), radical constructivism argues that it merely implies *viability*. As Glasersfeld (2002) noted, a key fits a lock not because it *resembles* the internal structure of the lock, but because it *operates effectively* within its constraints. Similarly, the weights of a neural network do not "match" the world; they represent viable solutions to the optimization problem defined by the loss function.

Where a metaphysical realist perspective, such as that of Doppelt (2007), might interpret "explanatory success" as necessitating a commitment to truth, operational constructivism views this success as the internal stabilization of a system's observations. Consequently, the convergence observed by Huh et al. (2024) can be understood as the result of massive selective pressure—gradient descent—pushing diverse architectures toward similar viable solutions for predicting human-generated output. From this perspective, the "miracle" is not the models' access to an objective world, but rather the profound statistical consistency of the sedimented communication they are trained on. Their convergence proves the stability of the social construction of reality, rather than the accessibility of an objective $Z$.

However, it is important to note that from a Luhmannian perspective, calling *operational stabilizations* "constructions" does not imply that they are unreal or mere illusions. On the contrary, they possess *operative reality*. For the system, reality is not a matter of matching an external world, but a matter of internal consistency and the recursive stabilization of its own operations. The convergence observed by Huh et al. (2024) therefore does not demonstrate an arrival at a Platonic $Z$, but rather the emergence of a highly *stable isomorphism* within the structural coupling of AI and society. From a Luhmannian perspective, this is not a "shared" reality in an ontological sense, but a *coordination of observations* mediated by the new connective medium of digital data (Baecker, 2025). In this view, $Z$ is not a territory we all inhabit, but a *consensual domain* (Maturana & Varela, 1980)—a statistical fixed point produced by the recursive operations of artificial and social systems alike.

## 6 What Follows: An Empirical Fork

A purely metaphysical disagreement risks being unfalsifiable on both sides, and we do not want to leave the argument there. Huh et al. (2024) do not stop at the hypothesis itself; they draw out its implications, and one implication in particular gives our account something concrete to disagree with. If representations are converging toward an increasingly accurate model of reality, they argue, then scaling should reduce hallucination: models conditioned on a sufficiently lossless and diverse set of measurements should increasingly say true things, because saying true things is what tracking $Z$ more closely amounts to. The same logic extends to bias: scaling should not remove bias, since bias enters through the data, but larger models should at least come to reflect the data's biases *accurately* rather than amplifying them, because accuracy with respect to the underlying statistical structure is what convergence delivers.

Our account agrees with half of this prediction and disagrees with the other half, and the disagreement is where it becomes testable. What scaling improves, on our reading, is not correspondence to reality but attunement to the *redundancy* of the training corpus—the degree to which a claim is stated, restated, and corroborated across many independent stretches of communication. A trivial machine trained under a shared code will, as argued in Section 5.1, lock onto whatever pattern is most consistently reinforced across its environment; high-redundancy claims—the kind that appear, in some form, across a large fraction of a training corpus—are exactly the pattern scale is best at reinforcing. This is not a different prediction from the PRH's for claims of this kind: both accounts expect hallucination on well-attested, frequently stated claims to fall with scale, though for different reasons (increasing correspondence to $Z$, on their account; increasing attunement to communicative redundancy, on ours).

The accounts diverge sharply, however, for claims that are true but rare: statements attested in only a small, idiosyncratic corner of the training corpus, or that run against a more frequently repeated but less accurate majority pattern. If convergence tracked reality, scale should still help here, since a bigger, more diverse sample of measurements should still make rare-but-true signal easier to recover, even if slowly. If convergence tracks redundancy, scale should not reliably help here at all, and could even hurt: a claim's statistical rarity, not its truth, is what determines how well a trivial machine locks onto it, so a confidently repeated falsehood that dominates the corpus should remain difficult to dislodge by scale alone, while a true but sparsely attested claim should remain vulnerable to being overwritten by whatever more redundant pattern surrounds it. There is already suggestive evidence on our side of this fork: Kandpal, Deng, Roberts, Wallace, and Raffel (2023) show that large language models systematically struggle to learn and recall long-tail factual knowledge—facts attested rarely in pretraining data—and that this gap closes only very slowly, if at all, with additional scale, in contrast to performance on well-attested facts, which improves reliably. This is precisely the bifurcation our account predicts and the PRH's does not: not a uniform reduction in hallucination as reality comes into focus, but a widening gap between what is well-attested and what is merely true.

The same fork applies to bias. The PRH predicts that scale makes a model's biases more accurately reflect the data's biases, treating this as a comparatively benign outcome, since "the data's biases" are at least a fact about the world the model is now representing faithfully. Our account predicts something narrower and less reassuring: scale reduces disagreement with whatever is most redundantly represented in the corpus, which includes but is not limited to bias in the ordinary sense—it also includes majority framings, dominant perspectives, and historically over-documented viewpoints, none of which need be more *true* than the minority positions they statistically outweigh. "More accurately reflecting the data's biases" and "more strongly attuned to the corpus's redundancy" sound similar, but they license different expectations about what further scaling and further data diversity should fix: the former suggests the residual problem is a data-quality problem, addressable by more representative sampling; the latter suggests it is a structural feature of any system that improves by locking onto redundancy, addressable, at best, only by changing what counts as redundant—which is a question about the composition of communication, not about a model's access to reality.

## 7 Conclusion

We have argued that the convergence Huh et al. (2024) document does not license the inference to a shared, mind-independent reality that the Platonic representation hypothesis draws from it. Operational constructivism denies the premise the inference needs: no system, biological, social, or artificial, has operational access to a world-in-itself, only to an environment already selected by its own distinctions. Applied to foundation models, this yields a claim stronger than the familiar objection that training data carries sociological bias, an objection Huh et al. (2024) themselves partly anticipate and treat as a matter of degree. The gap between representation and $Z$ is not a matter of degree; it is constitutive of what a structurally closed system is. We sharpened this further by asking whether large models are systems in Luhmann's strict sense at all, and argued that their very convergence is better explained by their being trivial, history-independent machines locking onto a shared statistical environment than by their being autopoietic systems discovering a world outside themselves—which yields a double deflation rather than a single one: convergence is evidence neither that models are approaching reality, against the PRH, nor that they are autonomously constructing a world of their own, against a careless reading of constructivism. What it is evidence of is models becoming increasingly well-adapted conduits for redistributing the redundancy already sedimented in human communication, a claim we tested directly against the PRH's own color-cooccurrence experiment and against its predictions about hallucination and bias under scale.

We opened by situating the Platonic representation hypothesis as the anchor case of a broader pattern rather than as an isolated target, and it is worth returning to that pattern briefly now that the argument is on the table. The "Anna Karenina scenario" (Bansal et al., 2021), in which well-performing networks are found to represent the world alike while poorly-performing ones fail idiosyncratically, invites exactly the same rereading we have given the PRH: what looks like convergence on a shared target is, on our account, better read as convergence on a shared code of success, which trivial machines under similar training pressure will satisfy in similar ways regardless of whether anything real stands behind it. Claims of *universality* in mechanistic interpretability—that independently trained vision models discover "the same" curve detectors and frequency detectors (Olah et al., 2020)—invite the same question we asked of CIELAB in Section 5.4: are these circuits converging on features of the physical world, or on regularities already imposed by the shared statistics of photographic images and the shared architecture of convolutional vision models, before either network is asked to do anything with them? And the neuroAI literature comparing artificial and biological visual representations (Schrimpf et al., 2018; Yamins et al., 2014) recruits evolution as a second independent witness to the same reality; our argument suggests that biological and artificial vision may instead be two different structurally coupled systems converging on the same redundancies in a shared, humanly structured visual environment, which would be a substantive finding in its own right, just not the one usually claimed for it. We do not attempt that fuller argument here—each of these literatures deserves the kind of direct, technical engagement we have given the PRH, not a hasty extension by analogy—but the PRH's willingness to state its metaphysics openly, with a name and a mathematical argument, is precisely what has let us show, in one well-developed case, what a systems-theoretical reading of the others would need to show as well.

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